

Quiz-1

Name: _____ Roll: _____ Date: 01/09/2025

Instructions

1. You are given 45 minutes to answer 11 questions.
2. Please read all questions carefully before attempting.
3. There are 10 MCQ, each having four options, with only one correct answer. Select the best possible answer.
4. Each correct response earns 2 marks and incorrect answer earns -1.
5. There is one subjective question, write your answer within the given space.

All The Best!

1. Which measure of central tendency is sensitive to extreme outliers?

☒ Mean	☐ Median	☐ Mode	☐ All are equally affected
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2. In one-hot encoding of a nominal attribute with 4 categories, how many binary columns are created?

☐ 3	☒ 4	☐ 5	☐ Depends on the dataset size
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3. Which of the following correctly describes the domain of Probability measure and random variables respectively?

☐ sample space and σ -algebra	☐ sample space	☐ σ -algebra	☐ sample space and σ -algebra
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4. A friend claims she can guess the suit of a card with $p > 0.25$, and you test with $n = 100$ trials. Which is correct?

☐ $H_0: p \geq 0.25, H_1: p < 0.25$	☒ $H_0: p \leq 0.25, H_1: p > 0.25$	☐ $H_0: p = 0.25, H_1: p \neq 0.25$	☐ $H_0: p > 0.25, H_1: p \leq 0.25$
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5. Which test is appropriate for comparing a sample mean to a known population mean when the population variance is known?

☐ t-test	☒ z-test	☐ χ^2 -test	☐ F-test
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6. In a dataset of T-shirts, the "colour" attribute has three possible values: red, blue, and green. How would this attribute be represented for a single "blue" T-shirt using one-hot encoding?

☐ [1]	☒ [0,1,0]	☐ [blue]	☐ [2]
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7. In hypothesis testing, the null hypothesis H_0 is

☐ Always False	☒ Assumed true until evidence suggests otherwise	☐ Research hypothesis	☐ Always True
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8. A histogram shows the distribution of retirement ages among professors. The x-axis represents age of retirement and the y-axis represents how many people retired. Based on typical career span (25 - 65), which of the following best describes the shape of this distribution?

☐ Positively Skewed	☒ Negatively Skewed	☐ Symmetric	☐ Cannot Say
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9. The Central Limit Theorem is a fundamental concept in statistics. It states that the distribution of the sum of a large number of independent random variables will tend to follow what distribution

☐ Uniform	☐ Exponential	☒ Normal	☐ Poisson Distribution
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10. In hypothesis test, Type – II error occurs when

☐ H_1 is rejected	☒ H_0 is not rejected	☐ H_0 is accepted	☐ H_1 is not accepted
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11. Which of the following correctly describes the domain of Probability measure and random variables respectively?

☒ σ -algebra and sample space	☐ sample space	☐ σ -algebra	☒ σ -algebra and sample space
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12. Which test is appropriate for comparing a sample mean to a known population mean when the population variance is not known?

☐ χ^2 -test	☒ t-test	☐ F-test	☐ z-test
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13. In hypothesis test, Type – I error occurs when

☒ H_0 is rejected	☐ H_0 is not accepted	☐ H_1 is not rejected	☐ H_1 is accepted
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14. You have a snakes and ladder board and you asked your friend to get a 4-sided unbiased die if she wants to play. She did not have such an die, instead she brought a biased coin. She said the probability of landing on Heads (H) is 0.4 and on Tails (T) is 0.6. As an expert, how would you design a procedure using only this coin to simulate a fair 4-sided die roll so that you two can play the game? (10 Marks)

The goal is to design a 4 sided virtual unbiased die, whose sides are 1, 2, 3 and 4.

We want $\Pr(1) = \Pr(2) = \Pr(3) = \Pr(4) = \frac{1}{4}$.

Notice that, $\Pr(\text{HTTT}) = \Pr(\text{THTT}) = \Pr(\text{TTHT}) = \Pr(\text{TTTH}) = 0.4 \cdot 0.6^3$.

Roll = 0

While not(Roll)

Toss the coin four times.

- If HTTT then Roll = 1 and return 1
- Elseif THTT then Roll = 1 and return 2
- Elseif TTHT then Roll = 1 and return 3
- Elseif TTTH then Roll = 1 and return 4

In the above steps the probability of getting 1, 2, 3, or 4 are equal, i.e., $\Pr(1) = \Pr(2) = \Pr(3) = \Pr(4) = \frac{0.4 \cdot 0.6^3}{4 \cdot 0.4 \cdot 0.6^3} = 0.25$.

Rubric:

- 2 marks for correctly identifying 4 different events but with equal probability.
- 2 marks for computing the probability of the 4 events.
- 4 marks for writing/explaining the steps e.g., as shown above.
- 2 marks for showing that the probability of virtual outcome is $\frac{1}{4}$.